

EMC filter

Installation and operating instructions

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(GB) Declaration of Conformity

We, Grundfos, declare under our sole responsibility that the product EMC filter, to which this declaration relates, is in conformity with these Council directives on the approximation of the laws of the EC member states:

- Low Voltage Directive (2006/95/EC).
Standard used: EN 60204.
- EMC Directive (2004/108/EC).
Standard used: EN 55011.

(D) Konformitätserklärung

Wir, Grundfos, erklären in alleiniger Verantwortung, dass das Produkt EMV-Filter, auf das sich diese Erklärung bezieht, mit den folgenden Richtlinien des Rates zur Angleichung der Rechtsvorschriften der EG-Mitgliedstaaten übereinstimmt:

- Niederspannungsrichtlinie (2006/95/EG).
Norm, die verwendet wurde: EN 60204.
- EMV-Richtlinie (2004/108/EG).
Norm, die verwendet wurde: EN 55011.

(DK) Overensstemmelseserklæring

Vi, Grundfos, erklærer under ansvar at produktet EMC-filter som denne erklæring omhandler, er i overensstemmelse med disse af Rådets direktiver om indbyrdes tilnærmelse til EF-medlemsstaternes lovgivning:

- Lavspændingsdirektivet (2006/95/EF).
Anvendt standard: EN 60204.
- EMC-direktivet (2004/108/EF).
Anvendt standard: EN 55011.

Bjerringbro, 15th December 2009



Svend Aage Kaae
Technical Director

EMC filter

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Warning

Prior to installation, read these installation and operating instructions. Installation and operation must comply with local regulations and accepted codes of good practice.

1. Application

An EMC filter is used for the purpose of reducing electrical noise. To meet EN 55011, limit B, Grundfos MMGE motors, 11-22 kW, require an EMC filter.

2. Before installation

Check that the filter kit contains:

- 1 EMC filter for max. 50 A contained in aluminium cabinet.

For connection to the motor terminal box:

- 1 reducer from M40 x 1.5 to M32 x 1.5
- 1 EMC nut M40 x 1.5
- 1 EMC cable gland M32 x 1.5
- installation and operating instructions.

3. Installation

3.1 Mounting of EMC filter

Mount the filter on an even surface. See sketch of installation principle in fig. 2.

The length and width of the cabinet are shown on the back of the cabinet. See dimensional sketch in fig. 1.

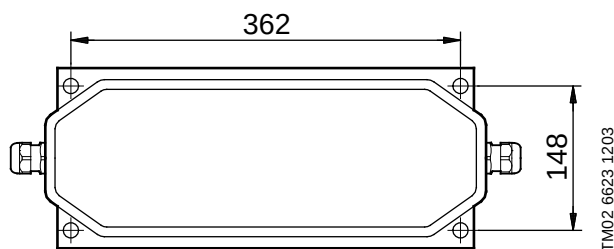


Fig. 1 Dimensional sketch

3.2 Electrical connection

The electrical connection must be carried out according to local regulations.

Warning

Never make any connections in the motor terminal box unless the power supply has been switched off for at least 5 minutes. Make sure that the power supply cannot be accidentally switched on.

The terminal box contains capacitors that have to discharge.

The motor may be hot for a while after being switched off.



3.2.1 Cables

Filter and motor must be connected with a screened cable. See fig. 2, pos. 1.

The ends of the screened cable must be terminated with EMC cable glands to ensure correct functioning of the filter. See sections 3.3 Connection of EMC filter and 3.4 Connection of EMC filter to motor terminal box.

The screened cable is not included in the filter kit.

Make sure that the screened cable has at least the same current value as the mains cable. See fig. 2, pos. 2.

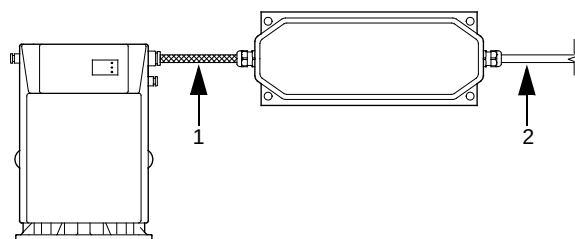
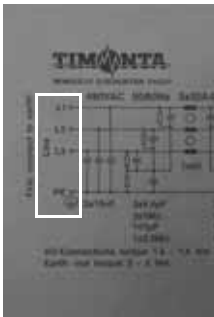
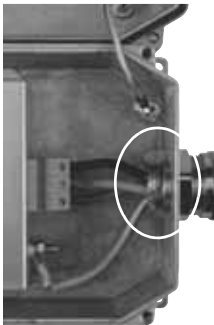
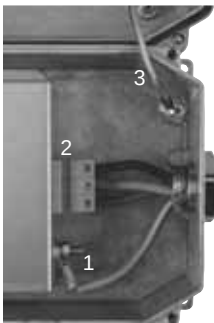
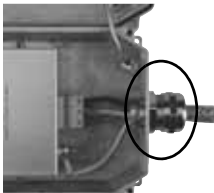


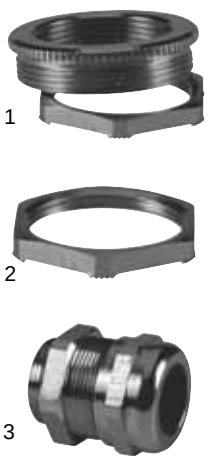
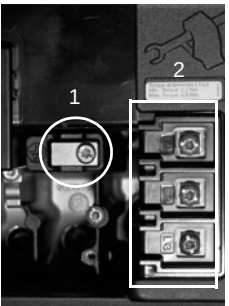
Fig. 2 Installation principle

TM02 6624 1203

3.3 Connection of EMC filter

Step	Action	Illustration
1.	- Remove the filter cover. - Connect the PE conductor to the earth terminal. - Connect the mains cable to terminals L1, L2 and L3 on the mains side of the filter. Details about mains connection are shown on the filter. See illustration. The mains side has no EMC cable gland.	 TM04 4616 1809
2.	- Strip about 60 mm of the outer sheath of the screened cable. - Pull back the screen 10-15 mm over the outer sheath. - Strip about 50 mm of the inner sheath.	 TM04 4615 1809
3.	- Insert the screened cable through the EMC cable gland fitted on the side of the filter facing the motor (the "motor side" of the filter). - Check that the screen makes good contact with the gland. The screen must be visible inside the gland.	 TM04 4612 1809
4.	- Connect the PE conductor (pos. 1) to the earth terminal. - Connect the conductors to terminals L1', L2' and L3' (pos. 2) in the motor side of the filter. - Check that the PE conductor between the cabinet cover and bottom (pos. 3) is fixed in position.	 TM04 4612 1809
5.	- Tighten the cable gland nut to secure the screened cable in position. This ensures good contact between screen and cabinet and provides strain-relief for the cable. - Fit the cover.	 TM04 4612 1809

3.4 Connection of EMC filter to motor terminal box

Step	Action	Illustration
1.	- Remove the terminal box cover. - Remove the existing cable gland from the side of the terminal box. - Screw the reducer with bush (pos. 1) on the outside of the terminal box. - Screw the EMC nut (pos. 2) on the inside of the terminal box. Make sure that the rough edges of the nut face the side of the terminal box to ensure good contact. - Screw the EMC cable gland (pos. 3) on the terminal box.	 TM04 4609 1809 - TM04 4608 1809 - TM04 4610 0809
2.	- Strip about 120 mm of the outer sheath of the screened cable. - Pull back the screen 10-15 mm over the outer sheath. - Strip about 110 mm of the inner sheath.	 TM04 4615 1809
3.	- Insert the screened cable through the EMC cable gland on the terminal box. - Check that the screen makes good contact with the gland. The screen must be visible inside the gland. - Connect the PE conductor to the earth terminal (pos. 1). - Connect the conductors to terminals L1, L2 and L3 (pos. 2). - Tighten the cable gland nut to secure the screened cable in position. This ensures good contact between screen and terminal box and provides strain-relief for the cable. - Fit the cover.	 TM04 6215 5109

4. Disposal

This product or parts of it must be disposed of in an environmentally sound way:

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.

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